12 The General Drift-Diffusion-Poisson Model

Introduction: From Equilibrium to Nonequilibrium Dynamics

Thus far, we have established the equilibrium carrier concentrations and derived the fundamental laws governing drift and diffusion currents. The next critical step is to transition from static equilibrium to **nonequilibrium dynamics**, which lies at the heart of all active electrical transport devices—from basic resistors to diodes, transistors, and integrated circuits. The primary objective of this section is to derive the fundamental set of coupled partial differential equations that describe how the electron and hole concentrations evolve as functions of spatial position $\mathbf{x} \in \mathbb{R}^3$ and time $t \in \mathbb{R}$. Once $n(\mathbf{x}, t)$ and $p(\mathbf{x}, t)$ are determined in response to an applied electric field $\mathcal{E}(\mathbf{x}, t)$, subject to specific geometric boundary conditions and material doping profiles, we can compute the terminal currents and ultimately extract the complete I–V characteristics of the device.

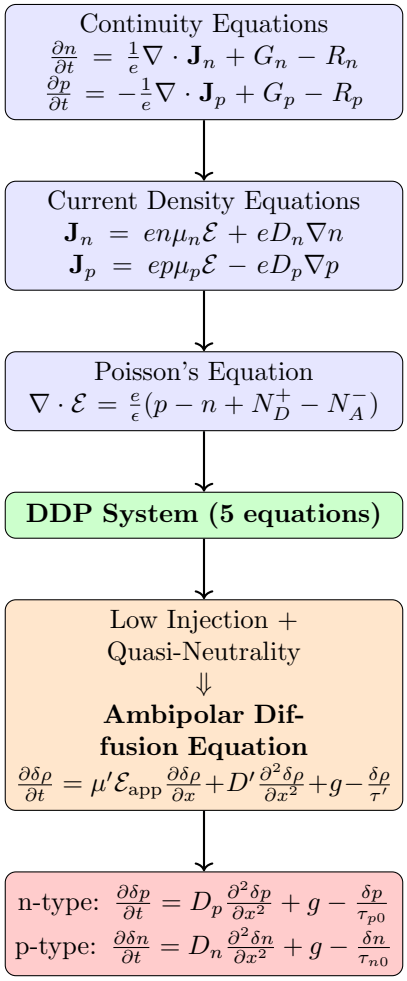


Figure 1: Roadmap from fundamental equations to the minority carrier diffusion equation.

12.1 Fundamental Equations Without Approximations

Definition of Physical Quantities

Below is a comprehensive list of the primary quantities used in the derivation. Emphasis is placed on their explicit space-time dependence where applicable.

Symbol	Physical Meaning	Units
$n(\mathbf{x},t)$	Electron concentration	cm^{-3}
$p(\mathbf{x},t)$	Hole concentration	cm^{-3}
$G_{\mathrm{n}}(\mathbf{x},t;n,p)$	Electron generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$R_{\mathrm{n}}(\mathbf{x},t;n,p)$	Electron recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$G_{\mathrm{p}}(\mathbf{x},t;n,p)$	Hole generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$R_{\mathrm{p}}(\mathbf{x},t;n,p)$	Hole recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$\mathbf{J}_{\mathrm{n}}(\mathbf{x},t)$	Total electron current density	$\mathrm{A\,cm}^{-2}$
$\mathbf{J}_{\mathrm{p}}(\mathbf{x},t)$	Total hole current density	$\mathrm{A\,cm}^{-2}$
e	Elementary charge (positive constant)	$1.602\times10^{-19}\,\mathrm{C}$
N_{D}^{+}	Ionized donor density (assumed fully ionized)	cm^{-3}
N_{A}^{-}	Ionized acceptor density (assumed fully ionized)	cm^{-3}
μ_{n}	Electron mobility	$\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$
μ_{p}	Hole mobility	$\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$
D_{n}	Electron diffusion coefficient	$\mathrm{cm}^2\,\mathrm{s}^{-1}$
D_{p}	Hole diffusion coefficient	$\mathrm{cm}^2\,\mathrm{s}^{-1}$
ϵ	Semiconductor permittivity	$\mathrm{F\,cm}^{-1}$

Recombination and Generation: Physical Context

We now consider the total recombination R and generation G rates, which encompass both equilibrium and nonequilibrium contributions. For the moment, we treat G as the total number of charge carriers generated per unit volume per unit time at location $\mathbf{x}$ and time t, given a local carrier content n and p. In general, G and R are highly nonlinear functionals of n and p, and may possess explicit space-time dependence $(\mathbf{x},t)$ beyond the implicit dependence carried by $n(\mathbf{x},t)$ and $p(\mathbf{x},t)$.

In semiconductor physics, we primarily focus on electron–hole pair recombination as the dominant loss mechanism in the bulk. Ohmic (Joule) dissipation is typically negligible in the semiconductor bulk and is instead accounted for at metal–semiconductor interfaces, ohmic contacts, or within metallic interconnects. Later in the course, we will introduce simplified phenomenological models (e.g., SRH, radiative, Auger) that effectively linearize or parameterize these relations for tractable analysis.

Derivation of the DDP Equations

The foundation of carrier transport is the **continuity equation**, which expresses local charge conservation. In its most general 3D form:

$$\frac{\partial\rho}{\partial t}+\nabla\cdot\mathbf{J}=\mathcal{G}-\mathcal{R}, \tag{6}$$

where ρ is the net charge density, $\mathbf{J}$ is the total current density, and $\mathcal{G}$, $\mathcal{R}$ are net generation and recombination source terms.

We now split this into two separate balance equations, one for electrons and one for holes. Using

$\rho_n = -en$ and $\rho_p = +ep$, and denoting the respective currents as $\mathbf{J}_\mathrm{n}$ and $\mathbf{J}_\mathrm{p}$, we obtain:

$$\frac{\partial n}{\partial t} = \frac{1}{e}\nabla \cdot \mathbf{J}_\mathrm{n} + G_\mathrm{n} - R_\mathrm{n}, \tag{7}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{e}\nabla \cdot \mathbf{J}_\mathrm{p} + G_\mathrm{p} - R_\mathrm{p}. \tag{8}$$

Note the sign difference: electrons carry negative charge, so their current divergence enters with a positive prefactor, whereas holes carry positive charge.

The total current densities are composed of **drift** (response to electric field) and **diffusion** (response to concentration gradients) components. Using the standard sign conventions:

$$\mathbf{J}_\mathrm{n}(\mathbf{x},t) = en\mu_\mathrm{n}\mathcal{E}(\mathbf{x},t) + eD_\mathrm{n}\nabla n, \tag{9}$$

$$\mathbf{J}_\mathrm{p}(\mathbf{x},t) = ep\mu_\mathrm{p}\mathcal{E}(\mathbf{x},t) - eD_\mathrm{p}\nabla p. \tag{10}$$

The final equation closes the system via **Gauss's Law** (Poisson equation), relating the electric field to the net charge density:

$$\nabla \cdot \mathcal{E}(\mathbf{x},t) = \frac{1}{\epsilon}\rho_\mathrm{net} = \frac{e}{\epsilon}\left[p(\mathbf{x},t) - n(\mathbf{x},t) + N_\mathrm{D}^+ - N_\mathrm{A}^-\right]. \tag{11}$$

The 3D Drift–Diffusion–Poisson (DDP) System

$$\boxed{\frac{\partial n}{\partial t} = \frac{1}{e}\nabla \cdot \mathbf{J}_\mathrm{n} + G_\mathrm{n} - R_\mathrm{n}} \tag{12}$$

$$\boxed{\frac{\partial p}{\partial t} = -\frac{1}{e}\nabla \cdot \mathbf{J}_\mathrm{p} + G_\mathrm{p} - R_\mathrm{p}} \tag{13}$$

$$\boxed{\mathbf{J}_\mathrm{n} = en\mu_\mathrm{n}\mathcal{E} + eD_\mathrm{n}\nabla n} \tag{14}$$

$$\boxed{\mathbf{J}_\mathrm{p} = ep\mu_\mathrm{p}\mathcal{E} - eD_\mathrm{p}\nabla p} \tag{15}$$

$$\boxed{\nabla \cdot \mathcal{E} = \frac{e}{\epsilon}\left(p - n + N_\mathrm{D}^+ - N_\mathrm{A}^-\right)} \tag{16}$$

One-Dimensional Reduction

For planar device structures or pedagogical analysis, we often assume all quantities vary only along a single spatial coordinate x. Replacing $\nabla \to \frac{\partial}{\partial x}\hat{\mathbf{x}}$ and $\mathcal{E}(\mathbf{x},t) \to \mathcal{E}(x,t)\,\hat{\mathbf{x}}$, the system reduces to:

The 1D Drift–Diffusion–Poisson System

$$\boxed{\frac{\partial n}{\partial t} = \frac{1}{e}\frac{\partial J_\mathrm{n}}{\partial x} + G_\mathrm{n} - R_\mathrm{n}} \tag{17}$$

$$\boxed{\frac{\partial p}{\partial t} = -\frac{1}{e}\frac{\partial J_\mathrm{p}}{\partial x} + G_\mathrm{p} - R_\mathrm{p}} \tag{18}$$

$$\boxed{J_\mathrm{n} = en\mu_\mathrm{n}\mathcal{E} + eD_\mathrm{n}\frac{\partial n}{\partial x}} \tag{19}$$

$$\boxed{J_\mathrm{p} = ep\mu_\mathrm{p}\mathcal{E} - eD_\mathrm{p}\frac{\partial p}{\partial x}} \tag{20}$$

$$\boxed{\frac{\partial \mathcal{E}}{\partial x} = \frac{e}{\epsilon}\left(p - n + N_\mathrm{D}^+ - N_\mathrm{A}^-\right)} \tag{21}$$

General Comments and Numerical Considerations

The five coupled partial differential equations above constitute the **Drift–Diffusion–Poisson (DDP) model**. They are highly nonlinear, strongly coupled, and generally lack closed-form analytical solutions. Even numerical resolution requires advanced computational techniques such as the Finite Element Method (FEM), Finite Difference Method (FDM), or specialized Semiconductor Device Simulators (TCAD tools like Sentaurus, Silvaco, or COMSOL).

The nonlinearity arises from:

- Products like $n\mathcal{E}$ and $p\mathcal{E}$ in the drift terms
- Spatial gradients ∇n, ∇p coupled to time derivatives
- Nonlinear, carrier-dependent $G(n,p)$ and $R(n,p)$ functions

Key Takeaways

- The DDP model provides a complete, self-consistent description of carrier transport and electrostatics in semiconductors.
- It is inherently **nonlinear** and **coupled**; analytical solutions require strategic approximations (depletion approximation, quasi-neutrality, steady-state, etc.).
- Modern device simulation relies on **FEM/FDM** discretization of these exact equations.
- In subsequent lectures, we will systematically simplify this system to derive tractable analytical models for diodes, BJTs, and MOSFETs while preserving the essential physics.

12.2 The Concept of Carrier Lifetime and Recombination Dynamics

In this subsection, we begin to simplify the highly coupled five-equation Drift–Diffusion–Poisson (DDP) system by introducing a series of well-justified physical assumptions. The first step in the analysis is to decompose the fundamental quantities into equilibrium (thermal) and nonequilibrium (excess) components.

Symbol	Physical Meaning	Units
n_0	Equilibrium electron concentration	cm^{-3}
p_0	Equilibrium hole concentration	cm^{-3}
$\delta n(\mathbf{x},t)$	Nonequilibrium (excess) electron concentration	cm^{-3}
$\delta p(\mathbf{x},t)$	Nonequilibrium (excess) hole concentration	cm^{-3}
$G_{\mathrm{th}}^{\mathrm{n}}$	Equilibrium electron generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$G_{\mathrm{th}}^{\mathrm{p}}$	Equilibrium hole generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$R_{\mathrm{th}}^{\mathrm{n}}$	Equilibrium electron recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$R_{\mathrm{th}}^{\mathrm{p}}$	Equilibrium hole recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$g_{\mathrm{n}}(\mathbf{x},t)$	Excess electron generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$g_{\mathrm{p}}(\mathbf{x},t)$	Excess hole generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$r_{\mathrm{n}}(\mathbf{x},t)$	Excess electron recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$r_{\mathrm{p}}(\mathbf{x},t)$	Excess hole recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$

The total generation and recombination rates are decomposed as:

$$G_{\rm n} = G_{\rm th}^{\rm n} + g_{\rm n}, \quad G_{\rm p} = G_{\rm th}^{\rm p} + g_{\rm p}, \tag{22}$$
$$R_{\rm n} = R_{\rm th}^{\rm n} + r_{\rm n}, \quad R_{\rm p} = R_{\rm th}^{\rm p} + r_{\rm p}. \tag{23}$$

At thermal equilibrium, we assume a homogeneous, steady-state distribution. Consequently, all spatial gradients and time derivatives of the equilibrium quantities vanish:

$$\frac{\partial n_0}{\partial t} = 0, \quad \frac{\partial p_0}{\partial t} = 0, \quad \nabla n_0 = 0, \quad \nabla p_0 = 0.$$

Furthermore, detailed balance requires that at equilibrium, every generation event is exactly compensated by a recombination event:

$$G_{\rm th}^{\rm n} = R_{\rm th}^{\rm n}, \qquad G_{\rm th}^{\rm p} = R_{\rm th}^{\rm p}.$$

This equality holds fundamentally because electron–hole (EH) pair creation via direct interband transitions (or thermal generation) occurs continuously. In steady state, the rate of pair creation must exactly match the rate of pair annihilation, otherwise carrier densities would grow or decay indefinitely.

Charge Neutrality: Global vs. Local It is critical to distinguish two distinct notions of charge neutrality.

- **Global Charge Neutrality**: The net total charge within a macroscopic, isolated semiconductor domain D is zero. Mathematically:

$$\int_D \rho(\mathbf{x}, t)\, {\rm d}V = 0 \quad \Rightarrow \quad \langle p - n + N_{\rm D}^+ - N_{\rm A}^- \rangle_D = 0.$$

  This condition ensures the device remains electrically neutral overall but does *not* imply $n(\mathbf{x}, t) = p(\mathbf{x}, t)$ at every point.

- **Local Charge Neutrality**: The net charge density vanishes at every point in the domain:

$$\rho(\mathbf{x}, t) = e\big[p(\mathbf{x}, t) - n(\mathbf{x}, t) + N_{\rm D}^+ - N_{\rm A}^-\big] = 0 \quad \forall \mathbf{x} \in D.$$

In equilibrium, we already know $p_0 - n_0 + N_{\rm D}^+ - N_{\rm A}^- = 0$. If we now impose local charge neutrality on the total nonequilibrium densities, we substitute $n = n_0 + \delta n$ and $p = p_0 + \delta p$:

$$(p_0 + \delta p) - (n_0 + \delta n) + N_{\rm D}^+ - N_{\rm A}^- = 0.$$

Canceling the equilibrium terms immediately yields a fundamental relation:

$$\boxed{\delta n(\mathbf{x}, t) = \delta p(\mathbf{x}, t)} \tag{24}$$

This local charge neutrality condition states that excess electrons and holes are generated (or removed) strictly in pairs. However, it is **crucial** to note that $\delta n = \delta p$ does *not* imply $n = p$. In n-type material, for example, $n_0 \gg p_0$, so even with equal excess densities, $n \gg p$ remains true.

Reduction of the Continuity Equations Assuming spatial homogeneity ($\nabla \cdot \mathbf{J}_{\rm n,p} = 0$) and applying the decomposition above, the continuity equations reduce to ordinary time derivatives. Since n_0 and p_0 are constant in time (${\rm d}n_0/{\rm d}t = {\rm d}p_0/{\rm d}t = 0$), we obtain:

$$\frac{{\rm d}\delta n}{{\rm d}t} = G_{\rm n} - R_{\rm n}, \tag{25}$$
$$\frac{{\rm d}\delta p}{{\rm d}t} = G_{\rm p} - R_{\rm p}. \tag{26}$$

These equations govern the temporal evolution of excess carriers but require a model for how $r_{\rm n,p}$ and $g_{\rm n,p}$ depend on carrier densities.

Modeling Recombination Dynamics The physics of recombination involves the random encounter of two distinct species: electrons and holes. This is a universal phenomenon observed in chemical kinetics, population dynamics, and laser physics. When modeling collision-driven interaction rates $R(n,p)$, several functional dependencies might be proposed: $R(n)$, $R(p)$, $R(n+p)$, or $R(np)$.

Physical counting arguments rule out the first three: a rate depending only on n or only on p contradicts the requirement that both species must participate in an encounter. A sum $n+p$ fails because the probability of pairwise collisions scales with the product of the population densities, not their sum. Thus, the only logically consistent leading-order dependence is:

$$R = \alpha np,$$

where α is a recombination coefficient. To account for possible asymmetries, we introduce distinct coefficients α_{n} and α_{p}. Expanding $R(n,p)$ in a Taylor series around equilibrium and retaining only the linear term (consistent with low-injection regimes) yields the standard bimolecular form.

At thermal equilibrium, $n_0 p_0 = n_i^2$, so $R_{\mathrm{th}} = \alpha_{\mathrm{n,p}} n_i^2$. Since $G_{\mathrm{th}}^{\mathrm{n,p}} = R_{\mathrm{th}}^{\mathrm{n,p}}$, we substitute into the rate equations:

$$\frac{\mathrm{d}n}{\mathrm{d}t} = \alpha_{\mathrm{n}} n_i^2 + g_{\mathrm{n}} - \alpha_{\mathrm{n}} np, \tag{27}$$

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \alpha_{\mathrm{p}} n_i^2 + g_{\mathrm{p}} - \alpha_{\mathrm{p}} np. \tag{28}$$

Why Focus on Minority Carriers?

In practical device analysis, we typically derive dynamic equations for **minority carriers** because:

- Minority carrier concentrations are orders of magnitude smaller than majority carriers, making their *relative* change much larger and more sensitive to external perturbations.
- Diffusion currents in p-n junctions, BJTs, and solar cells are dominated by minority carrier gradients.
- Majority carriers rapidly screen external fields and equilibrate locally, while minority carriers dictate the temporal response speed (switching, modulation, photoconductive decay).

Thus, tracking minority carrier dynamics captures the essential transient and steady-state behavior of active semiconductor devices.

Let us focus on an **n-type** material and derive the equation for minority carriers (holes). Using $n = n_0 + \delta n$, $p = p_0 + \delta p$, and the local neutrality condition $\delta n = \delta p$, the excess recombination rate for holes becomes:

$$r_{\mathrm{p}} = R_{\mathrm{p}} - R_{\mathrm{th}}^{\mathrm{p}} = \alpha_{\mathrm{p}}(n_0 + \delta p)(p_0 + \delta p) - \alpha_{\mathrm{p}} n_i^2.$$

Since $n_0 p_0 = n_i^2$, the equilibrium terms cancel exactly, leaving:

$$r_{\mathrm{p}} = \alpha_{\mathrm{p}}\left[n_0 \delta p + p_0 \delta p + (\delta p)^2\right] = \alpha_{\mathrm{p}}\left[(n_0 + p_0)\delta p + (\delta p)^2\right].$$

The exact continuity equation for excess holes is therefore:

$$\frac{\mathrm{d}\delta p}{\mathrm{d}t} = g_{\mathrm{p}} - \alpha_{\mathrm{p}}(n_0 + p_0)\delta p - \alpha_{\mathrm{p}}(\delta p)^2.$$

This equation is nonlinear due to the $(\delta p)^2$ term and generally requires numerical solution. To obtain an analytical model, we invoke the **low-injection approximation**:

$$\boxed{\delta n = \delta p \ll n_0 + p_0.}$$

Under this condition, the quadratic term is negligible compared to the linear term. The equation linearizes to:

$$\frac{\mathrm{d}\delta p}{\mathrm{d}t} = g_{\mathrm{p}} - \frac{\delta p}{\tau_{\mathrm{p0}}},$$

where

$$\boxed{\tau_{\mathrm{p0}} = \frac{1}{\alpha_{\mathrm{p}}(n_0 + p_0)} \approx \frac{1}{\alpha_{\mathrm{p}} n_0}}$$

The parameter τ_{p0} is called the **minority carrier lifetime**. To see its physical meaning, consider a pulse excitation where $g_{\mathrm{p}} = 0$ for $t > 0$ and an initial excess density $\delta p(0) = \delta p_0$. The solution is:

$$\delta p(t) = \delta p_0 \, e^{-t/\tau_{\mathrm{p0}}}.$$

When $t \gg \tau_{\mathrm{p0}}$, the excess pulse effectively disappears. Thus, τ_{p0} represents the average time an excess hole survives in the semiconductor before recombining with an electron.

A completely symmetric derivation for **p-type** material yields the electron lifetime:

$$\boxed{\tau_{\mathrm{n0}} = \frac{1}{\alpha_{\mathrm{n}}(n_0 + p_0)} \approx \frac{1}{\alpha_{\mathrm{n}} p_0}}$$

Lifetime vs. Momentum Relaxation Time

It is vital **not** to confuse the carrier lifetimes $\tau_{\mathrm{n0}}, \tau_{\mathrm{p0}}$ with the momentum relaxation (scattering) times $\tau_{\mathrm{n}}, \tau_{\mathrm{p}}$ introduced in the drift analysis. They describe fundamentally different physical processes:

- $\tau_{\mathrm{n}}, \tau_{\mathrm{p}}$: Average time between momentum-randomizing collisions with the lattice or impurities (picosecond scale).
- $\tau_{\mathrm{n0}}, \tau_{\mathrm{p0}}$: Average time before an electron and hole meet and annihilate (microsecond to millisecond scale).

Typical values at room temperature are summarized below:

Material	τ_{scat} (s)	τ_{n0} (s)	τ_{p0} (s)	$\tau_{\mathrm{life}}/\tau_{\mathrm{scat}}$
Silicon (Si)	$\sim 10^{-13}$	10^{-5}–10^{-3}	10^{-5}–10^{-3}	$\sim 10^{8}$–10^{10}
Gallium Arsenide (GaAs)	$\sim 10^{-13}$	10^{-9}–10^{-7}	10^{-9}–10^{-7}	$\sim 10^{4}$–10^{6}
Germanium (Ge)	$\sim 10^{-13}$	10^{-4}–10^{-2}	10^{-4}–10^{-2}	$\sim 10^{9}$–10^{11}

The carrier lifetime is typically *orders of magnitude larger* than the relaxation time. This is physically necessary: a carrier must scatter off the lattice or impurities many thousands of times before it randomly encounters a carrier of the opposite sign to annihilate in an EH recombination event. If $\tau_{\mathrm{life}} \approx \tau_{\mathrm{scat}}$, conduction would be impossible because all carriers would recombine immediately upon generation.

Finally, we express the nonequilibrium recombination rates in the standard linear form:

$$\boxed{r_{\mathrm{n}} = \frac{\delta n}{\tau_{\mathrm{n0}}}, \qquad r_{\mathrm{p}} = \frac{\delta p}{\tau_{\mathrm{p0}}}.}$$

These represent the canonical recombination models used throughout this course. They close the DDP system under the low-injection, local charge neutrality, and homogeneous assumptions, enabling tractable analytical solutions for diode, transistor, and optoelectronic device characteristics.

12.3 Derivation of Coupled Carrier Density Equations

In the previous section, we assumed spatial homogeneity to reduce the transport problem to ordinary differential equations. We now relax this assumption to treat the general case where carrier concentrations vary in both space and time. The objective is to derive two coupled partial differential equations governing the dynamics of the excess carrier densities $\delta n(\mathbf{x}, t)$ and $\delta p(\mathbf{x}, t)$.

We begin by decomposing the total generation and recombination rates into equilibrium and excess components:

$$G_{\mathrm{n}} - R_{\mathrm{n}} = g_{\mathrm{n}} - r_{\mathrm{n}}, \qquad G_{\mathrm{p}} - R_{\mathrm{p}} = g_{\mathrm{p}} - r_{\mathrm{p}}.$$

Since the equilibrium concentrations n_0 and p_0 are spatially uniform and time-independent, the partial derivatives satisfy

$$\frac{\partial n}{\partial t} = \frac{\partial \delta n}{\partial t}, \quad \frac{\partial p}{\partial t} = \frac{\partial \delta p}{\partial t}, \quad \nabla n = \nabla \delta n, \quad \nabla p = \nabla \delta p.$$

Substituting these into the general continuity equations yields:

$$\frac{\partial \delta n}{\partial t} = \frac{1}{e}\nabla \cdot \mathbf{J}_{\mathrm{n}} + g_{\mathrm{n}} - r_{\mathrm{n}}, \tag{29}$$

$$\frac{\partial \delta p}{\partial t} = -\frac{1}{e}\nabla \cdot \mathbf{J}_{\mathrm{p}} + g_{\mathrm{p}} - r_{\mathrm{p}}. \tag{30}$$

Next, we substitute the Drift–Diffusion–Poisson (DDP) expressions for the current densities:

$$\mathbf{J}_{\mathrm{n}} = en\mu_{\mathrm{n}}\mathcal{E} + eD_{\mathrm{n}}\nabla n, \qquad \mathbf{J}_{\mathrm{p}} = ep\mu_{\mathrm{p}}\mathcal{E} - eD_{\mathrm{p}}\nabla p.$$

Inserting these into (29) and (30) gives:

$$\frac{\partial \delta n}{\partial t} = \frac{1}{e}\nabla \cdot (en\mu_{\mathrm{n}}\mathcal{E} + eD_{\mathrm{n}}\nabla n) + g_{\mathrm{n}} - r_{\mathrm{n}}, \tag{31}$$

$$\frac{\partial \delta p}{\partial t} = -\frac{1}{e}\nabla \cdot (ep\mu_{\mathrm{p}}\mathcal{E} - eD_{\mathrm{p}}\nabla p) + g_{\mathrm{p}} - r_{\mathrm{p}}. \tag{32}$$

Assuming constant mobility and diffusion coefficients, we extract them from the divergence operator:

$$\frac{\partial \delta n}{\partial t} = \mu_{\mathrm{n}}\nabla \cdot (n\mathcal{E}) + D_{\mathrm{n}}\nabla^2 n + g_{\mathrm{n}} - r_{\mathrm{n}}, \tag{33}$$

$$\frac{\partial \delta p}{\partial t} = -\mu_{\mathrm{p}}\nabla \cdot (p\mathcal{E}) + D_{\mathrm{p}}\nabla^2 p + g_{\mathrm{p}} - r_{\mathrm{p}}. \tag{34}$$

We now apply the **Leibniz product rule** for the divergence of a scalar field ψ multiplied by a vector field $\mathbf{A}$:

$$\nabla \cdot (\psi \mathbf{A}) = \psi\, \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla \psi.$$

Applying this rule term-by-term and substituting $n = n_0 + \delta n$, $p = p_0 + \delta p$ (with $\nabla n_0 = \nabla p_0 = 0$), we obtain the step-by-step expansions:

$$\begin{aligned}
\frac{\partial \delta n}{\partial t} &= \mu_{\mathrm{n}}\nabla \cdot (n\mathcal{E}) + D_{\mathrm{n}}\nabla^2 n + g_{\mathrm{n}} - r_{\mathrm{n}} \\
&= \mu_{\mathrm{n}}\left[n\nabla \cdot \mathcal{E} + \mathcal{E} \cdot \nabla n\right] + D_{\mathrm{n}}\nabla^2 n + g_{\mathrm{n}} - r_{\mathrm{n}} \\
&= \mu_{\mathrm{n}}(n_0 + \delta n)\nabla \cdot \mathcal{E} + \mu_{\mathrm{n}}\mathcal{E} \cdot \nabla \delta n + D_{\mathrm{n}}\nabla^2 \delta n + g_{\mathrm{n}} - r_{\mathrm{n}},
\end{aligned} \tag{35}$$

$$\begin{aligned}
\frac{\partial \delta p}{\partial t} &= -\mu_{\mathrm{p}}\nabla \cdot (p\mathcal{E}) + D_{\mathrm{p}}\nabla^2 p + g_{\mathrm{p}} - r_{\mathrm{p}} \\
&= -\mu_{\mathrm{p}}\left[p\nabla \cdot \mathcal{E} + \mathcal{E} \cdot \nabla p\right] + D_{\mathrm{p}}\nabla^2 p + g_{\mathrm{p}} - r_{\mathrm{p}} \\
&= -\mu_{\mathrm{p}}(p_0 + \delta p)\nabla \cdot \mathcal{E} - \mu_{\mathrm{p}}\mathcal{E} \cdot \nabla \delta p + D_{\mathrm{p}}\nabla^2 \delta p + g_{\mathrm{p}} - r_{\mathrm{p}}.
\end{aligned} \tag{36}$$

Equations (35) and (36) are second-order partial differential equations in space and explicitly involve $\nabla \cdot \mathcal{E}$. To close the system, we invoke Gauss's Law:

$$\nabla \cdot \mathcal{E} = \frac{e}{\epsilon}\left(p - n + N_{\mathrm{D}}^{+} - N_{\mathrm{A}}^{-}\right).$$

Using the equilibrium charge neutrality condition $p_0 - n_0 + N_{\mathrm{D}}^{+} - N_{\mathrm{A}}^{-} = 0$, we substitute $p = p_0 + \delta p$ and $n = n_0 + \delta n$ to obtain:

$$\nabla \cdot \mathcal{E} = \frac{e}{\epsilon}\left(\delta p - \delta n\right). \tag{37}$$

Note that we have **not** assumed local charge neutrality ($\delta n = \delta p$) at this stage. Retaining $\nabla \cdot \mathcal{E}$ explicitly allows us to treat the most general situation rigorously. (The consequences of imposing local charge neutrality will be examined in detail when we discuss the ambipolar approximation in subsequent lectures.)

Self-Consistent Coupling

The system (35)–(37) is inherently **nonlinear and strongly coupled**. Physically:

- Spatial gradients or temporal changes in δn and δp alter the net charge density.
- This modified charge density generates an electric field $\mathcal{E}$ via (37).
- The resulting $\mathcal{E}$ feeds back into the drift terms $\mu\mathcal{E} \cdot \nabla \delta n$ and $\mu\mathcal{E} \cdot \nabla \delta p$, which in turn modify the carrier evolution.

Consequently, these equations must be solved **self-consistently**. In analytical treatments, simplifying assumptions are introduced. In computational device simulation, iterative numerical schemes (e.g., Gummel iteration or Newton–Raphson coupled with FEM/FDM) are employed to converge to the self-consistent $\delta n(\mathbf{x}, t)$, $\delta p(\mathbf{x}, t)$, and $\mathcal{E}(\mathbf{x}, t)$ fields at each time step.

For planar device geometries, we reduce the system to one spatial dimension (x). Replacing $\nabla \to \frac{\partial}{\partial x}$, $\nabla^2 \to \frac{\partial^2}{\partial x^2}$, and $\mathcal{E} \cdot \nabla \to \mathcal{E}\frac{\partial}{\partial x}$, we obtain:

$$\boxed{\frac{\partial \delta n}{\partial t} = \mu_{\mathrm{n}}\mathcal{E}\frac{\partial \delta n}{\partial x} + \mu_{\mathrm{n}}(n_0 + \delta n)\frac{e}{\epsilon}(\delta p - \delta n) + D_{\mathrm{n}}\frac{\partial^2 \delta n}{\partial x^2} + g_{\mathrm{n}} - r_{\mathrm{n}}} \tag{38}$$

$$\boxed{\frac{\partial \delta p}{\partial t} = -\mu_{\mathrm{p}}\mathcal{E}\frac{\partial \delta p}{\partial x} - \mu_{\mathrm{p}}(p_0 + \delta p)\frac{e}{\epsilon}(\delta p - \delta n) + D_{\mathrm{p}}\frac{\partial^2 \delta p}{\partial x^2} + g_{\mathrm{p}} - r_{\mathrm{p}}} \tag{39}$$

Finally, under the **low-injection regime** ($\delta n = \delta p \ll n_0 + p_0$), the recombination rates linearize to:

$$r_{\mathrm{n}} = \frac{\delta n}{\tau_{\mathrm{n0}}}, \qquad r_{\mathrm{p}} = \frac{\delta p}{\tau_{\mathrm{p0}}},$$

where τ_{n0} and τ_{p0} are the minority carrier lifetimes derived previously. Substituting these into the 1D equations yields the standard coupled transport equations used in most analytical device models:

$$\frac{\partial \delta n}{\partial t} = \mu_{\mathrm{n}}\mathcal{E}\frac{\partial \delta n}{\partial x} + \mu_{\mathrm{n}}(n_0 + \delta n)\frac{e}{\epsilon}(\delta p - \delta n) + D_{\mathrm{n}}\frac{\partial^2 \delta n}{\partial x^2} + g_{\mathrm{n}} - \frac{\delta n}{\tau_{\mathrm{n0}}}, \tag{40}$$

$$\frac{\partial \delta p}{\partial t} = -\mu_{\mathrm{p}}\mathcal{E}\frac{\partial \delta p}{\partial x} - \mu_{\mathrm{p}}(p_0 + \delta p)\frac{e}{\epsilon}(\delta p - \delta n) + D_{\mathrm{p}}\frac{\partial^2 \delta p}{\partial x^2} + g_{\mathrm{p}} - \frac{\delta p}{\tau_{\mathrm{p0}}}. \tag{41}$$

These equations form the complete dynamical foundation for analyzing carrier injection, extraction, and diffusion in semiconductor devices under realistic spatially varying conditions.

Key Features of the Coupled Transport System

- **Second-Order Spatial Dependence**: The Laplacian terms $D_{\mathrm{n}}\partial^2\delta n/\partial x^2$ and $D_{\mathrm{p}}\partial^2\delta p/\partial x^2$ capture diffusion-driven carrier spreading from regions of high concentration to low concentration.
- **Field-Carrier Coupling**: The terms proportional to $\mathcal{E}\,\partial\delta n/\partial x$ and the explicit $(\delta p - \delta n)$ factors demonstrate how spatial charge imbalances instantly modify the local electric field, which in turn feeds back into the drift velocities.
- **Linearized Recombination Dynamics**: Under low injection, the recombination channels reduce to simple first-order decay terms $-\delta n/\tau_{\mathrm{n0}}$ and $-\delta p/\tau_{\mathrm{p0}}$, introducing characteristic relaxation timescales that compete with transport.
- **Boundary Conditions**: To obtain unique physical solutions, these PDEs must be supplemented with appropriate boundary conditions at device interfaces (e.g., ohmic contacts, surface recombination velocities, or reflective boundaries).

The coupled system above constitutes the most general description of nonequilibrium carrier transport within the drift-diffusion framework. While analytically formidable due to the explicit nonlinear coupling between δn, δp, and $\mathcal{E}$, this formulation rigorously captures the competition between injection, diffusion, drift, and recombination. In practical device simulation, these equations are discretized and solved iteratively using self-consistent numerical solvers (e.g., Gummel or Newton–Raphson methods) to resolve the electrostatic and transport fields at each spatiotemporal node.

The Inconsistency of Exact Local Charge Neutrality

Motivational Scenario: Uniform Optical Excitation Consider a semiconductor sample uniformly illuminated by light with photon energy $h\nu > E_g$. Each absorbed photon generates one electron-hole pair. Therefore, in the bulk of the semiconductor (away from surfaces and contacts), the generation rates for electrons and holes are intrinsically equal:

$$g_n(\mathbf{x},t) = g_p(\mathbf{x},t) \equiv g(\mathbf{x},t)$$

This is a fundamental physical fact: photons create electrons and holes in pairs. There is no mechanism in the bulk that generates an electron without simultaneously generating a hole, or vice versa.

Now, in the low-injection regime, it is tempting to also enforce **exact local charge neutrality** everywhere by setting $\delta n(x,t) = \delta p(x,t) \equiv \delta\rho(x,t)$. This assumption is often motivated by the observation that bulk semiconductors cannot sustain large space-charge regions over macroscopic distances — any local imbalance would be quickly neutralized by the high conductivity of the majority carriers.

Under these two assumptions — equal generation rates and equal excess carrier densities — the system appears to simplify dramatically. Substituting these conditions into the general 1D coupled carrier density equations, the Poisson space-charge term vanishes identically:

$$\frac{e}{\epsilon}(\delta p - \delta n) = 0.$$

The two evolution equations then decouple from the electric field and reduce to:

$$\frac{\partial\delta\rho}{\partial t} = \mu_{\mathrm{n}}\mathcal{E}\frac{\partial\delta\rho}{\partial x} + D_{\mathrm{n}}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{n0}}}, \tag{42}$$

$$\frac{\partial\delta\rho}{\partial t} = -\mu_{\mathrm{p}}\mathcal{E}\frac{\partial\delta\rho}{\partial x} + D_{\mathrm{p}}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{p0}}}. \tag{43}$$

The Inconsistency Since both equations describe the same physical quantity $\delta\rho(x,t)$, their left-hand sides are identical. For consistency, the right-hand sides must also be equal at every point (x,t). Equating (42) and (43) yields the constraint:

$$(\mu_{\mathrm{n}}+\mu_{\mathrm{p}})\mathcal{E}\frac{\partial\delta\rho}{\partial x}+(D_{\mathrm{n}}-D_{\mathrm{p}})\frac{\partial^2\delta\rho}{\partial x^2}+\delta\rho\left(\frac{1}{\tau_{\mathrm{p0}}}-\frac{1}{\tau_{\mathrm{n0}}}\right)=0.$$

For this equation to hold **identically** for all possible $\delta\rho(x,t)$ and $\mathcal{E}(x,t)$ that might arise in a device, the coefficients would need to satisfy:

- $\mu_{\mathrm{n}}+\mu_{\mathrm{p}}=0 \quad\Rightarrow\quad \mu_{\mathrm{n}}=-\mu_{\mathrm{p}}$ (impossible, since both mobilities are strictly positive),
- $D_{\mathrm{n}}-D_{\mathrm{p}}=0 \quad\Rightarrow\quad D_{\mathrm{n}}=D_{\mathrm{p}}$ (generally false, as $D=\mu k_B T/e$ and $\mu_{\mathrm{n}}\neq\mu_{\mathrm{p}}$),
- $\frac{1}{\tau_{\mathrm{p0}}}-\frac{1}{\tau_{\mathrm{n0}}}=0 \quad\Rightarrow\quad \tau_{\mathrm{n0}}=\tau_{\mathrm{p0}}$ (only true in highly symmetric or compensated materials).

Since these equalities are not satisfied in real semiconductors, the system (42)–(43) is **mathematically inconsistent**.

Physical Interpretation: Why Exact Neutrality Fails

The contradiction reveals a fundamental dynamical fact: electrons and holes possess different mobilities and diffusion coefficients. If they were forced to move exactly together ($\delta n=\delta p$ everywhere), their natural drift and diffusion velocities would differ. This mismatch would instantly generate a local separation of charge, which in turn produces an internal electric field.
Therefore, **exact local charge neutrality cannot be strictly enforced** in the presence of spatial gradients or external fields. The semiconductor maintains a *quasi-neutral* state where $\delta n\approx\delta p$, but a tiny, self-consistent deviation $\delta p-\delta n\neq 0$ is precisely what generates the internal field required to couple the electron and hole flows.

Assumption	Physical Motivation	Why It Fails
$g_n=g_p$	Optical generation creates e-h pairs	✓ Correct in bulk, not the problem
$\delta n=\delta p$	Bulk cannot sustain large space charge	× Fails because $\mu_n\neq\mu_p$, $D_n\neq D_p$

How Do We Proceed? The Quasi-Neutrality and Ambipolar Approximations The resolution lies in relaxing the strict equality $\delta n=\delta p$ while retaining the physical insight that bulk semiconductors cannot sustain large space-charge regions outside depletion zones. We adopt the **quasi-neutrality assumption**:

$$\boxed{\delta n(x,t)\approx\delta p(x,t),\quad \text{but}\quad \nabla\cdot\mathcal{E}\neq 0\ \text{(small but finite)}.}$$

Instead of discarding the Poisson term, we use it to eliminate the unknown electric field $\mathcal{E}$ from the coupled system. By combining the two continuity equations in a way that cancels the drift-diffusion mismatch, we can derive a single transport equation for the excess carrier density. This procedure yields the celebrated **ambipolar diffusion equation**, which introduces effective transport parameters (ambipolar mobility μ' and ambipolar diffusion coefficient D') that describe the coupled motion of electrons and holes as a single quasi-particle ensemble.

In the following subsection, we will rigorously derive this ambipolar transport equation, show how it resolves the inconsistency highlighted above, and demonstrate its power in analyzing bipolar devices such as p-n junctions, solar cells, and photodetectors.

12.4 The Ambipolar Approximation

The strong electrostatic coupling between electrons and holes often enforces quasi-neutrality in the bulk of optoelectronic and bipolar devices. In this subsection, we will exploit this property to combine the two continuity equations into a single *ambipolar diffusion equation*, dramatically simplifying analytical treatments of p-n junctions, photodetectors, and bipolar transistors while preserving the essential physics of coupled carrier transport.

Key Physical Idea: The Internal Electric Field

When excess electrons and holes are injected into a semiconductor, they tend to move under the influence of:

- Any **applied electric field** $\mathcal{E}_{\text{app}}$ (from external bias)
- **Diffusion** due to concentration gradients

However, electrons and holes have different mobilities ($\mu_{\text{n}} \neq \mu_{\text{p}}$) and different diffusion coefficients ($D_{\text{n}} \neq D_{\text{p}}$). If they were to move independently, their natural drift and diffusion velocities would differ. This would instantly create a local separation of charge, generating an **internal electric field** $\mathcal{E}_{\text{int}}$ via Poisson's equation.

The total electric field is therefore:

$$\boxed{\mathcal{E}_{\text{total}} = \mathcal{E}_{\text{app}} + \mathcal{E}_{\text{int}}}$$

This internal field acts as a **restoring force** that binds the excess electron and hole populations together. Physically:

- If electrons (negative) try to move faster than holes (positive), the resulting charge imbalance creates an internal field that **slows down** the faster species and **accelerates** the slower species
- This Coulomb coupling ensures that the excess electron and hole pulses travel together as a single *ambipolar* packet

Crucial Note: Binding Only Applies to Excess Carriers

The internal field binds the **excess** carriers (δn and δp) to each other. It does **not** bind the total carrier populations ($n = n_0 + \delta n$, $p = p_0 + \delta p$).
For example, in n-type material:

- The majority carrier concentration n_0 is much larger than δn
- The internal field does not prevent the majority electrons from flowing independently to contacts
- This is why we can have large terminal currents while maintaining $\delta n \approx \delta p$ in the bulk

Derivation of the Ambipolar Transport Equation

We now present the complete derivation of the ambipolar transport equation. Starting from the 1D continuity equations for excess carriers under low injection ($\delta n = \delta p = \delta\rho$) and assuming quasi-neutrality, we combine the equations to eliminate the electric field.

The 1D continuity equations for electrons and holes under low injection, including drift, diffusion, generation, and recombination, are:

$$\frac{\partial \delta n}{\partial t} = \mu_{\text{n}} \mathcal{E}_{\text{total}} \frac{\partial \delta n}{\partial x} + D_{\text{n}} \frac{\partial^2 \delta n}{\partial x^2} + g - \frac{\delta n}{\tau_{\text{n0}}}, \tag{44}$$

$$\frac{\partial \delta p}{\partial t} = -\mu_{\text{p}} \mathcal{E}_{\text{total}} \frac{\partial \delta p}{\partial x} + D_{\text{p}} \frac{\partial^2 \delta p}{\partial x^2} + g - \frac{\delta p}{\tau_{\text{p0}}}. \tag{45}$$

Under quasi-neutrality, we set $\delta n = \delta p = \delta\rho$. Also, the total electric field $\mathcal{E}_{\text{total}}$ is the sum of applied and internal fields. However, for the purpose of deriving the ambipolar equation, we treat $\mathcal{E}_{\text{total}}$ as the field that appears in both equations.

To eliminate $\mathcal{E}_{\text{total}}$, we multiply Equation (44) by $\mu_{\text{p}}p_0$ and Equation (45) by $\mu_{\text{n}}n_0$, then add them. This weighting choice combines the drift terms into a single effective drift term characterized by the ambipolar mobility μ'.

First, compute the weighted equations:

$$\mu_{\text{p}}p_0\frac{\partial\delta\rho}{\partial t} = \mu_{\text{n}}\mu_{\text{p}}p_0\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x} + \mu_{\text{p}}p_0D_{\text{n}}\frac{\partial^2\delta\rho}{\partial x^2} + \mu_{\text{p}}p_0g - \mu_{\text{p}}p_0\frac{\delta\rho}{\tau_{\text{n0}}}, \tag{46}$$

$$\mu_{\text{n}}n_0\frac{\partial\delta\rho}{\partial t} = -\mu_{\text{n}}\mu_{\text{p}}n_0\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x} + \mu_{\text{n}}n_0D_{\text{p}}\frac{\partial^2\delta\rho}{\partial x^2} + \mu_{\text{n}}n_0g - \mu_{\text{n}}n_0\frac{\delta\rho}{\tau_{\text{p0}}}. \tag{47}$$

Now add Equations (46) and (47):

$$(\mu_{\text{p}}p_0+\mu_{\text{n}}n_0)\frac{\partial\delta\rho}{\partial t} = \mu_{\text{n}}\mu_{\text{p}}(p_0-n_0)\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x}+(\mu_{\text{p}}p_0D_{\text{n}}+\mu_{\text{n}}n_0D_{\text{p}})\frac{\partial^2\delta\rho}{\partial x^2}+(\mu_{\text{p}}p_0+\mu_{\text{n}}n_0)g-\left(\frac{\mu_{\text{p}}p_0}{\tau_{\text{n0}}}+\frac{\mu_{\text{n}}n_0}{\tau_{\text{p0}}}\right)\delta\rho.$$

The drift term simplifies to:

$$-\mu_{\text{n}}\mu_{\text{p}}(n_0-p_0)\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x}.$$

Divide both sides by $(\mu_{\text{p}}p_0+\mu_{\text{n}}n_0)$:

$$\frac{\partial\delta\rho}{\partial t} = -\frac{\mu_{\text{n}}\mu_{\text{p}}(n_0-p_0)}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x} + \frac{\mu_{\text{p}}p_0D_{\text{n}}+\mu_{\text{n}}n_0D_{\text{p}}}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\mu_{\text{p}}p_0/\tau_{\text{n0}}+\mu_{\text{n}}n_0/\tau_{\text{p0}}}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}\delta\rho.$$

Thus, we identify the **ambipolar mobility** μ' and **ambipolar diffusion coefficient** D':

$$\boxed{\mu' = \frac{\mu_{\text{n}}\mu_{\text{p}}(n_0-p_0)}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}}$$

$$\boxed{D' = \frac{\mu_{\text{p}}p_0D_{\text{n}}+\mu_{\text{n}}n_0D_{\text{p}}}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}}$$

Using the Einstein relation $D = \mu k_BT/e$, we substitute $\mu_{\text{n}} = \frac{e}{k_BT}D_{\text{n}}$ and $\mu_{\text{p}} = \frac{e}{k_BT}D_{\text{p}}$ into the expression for D'. The constant factors $\frac{e}{k_BT}$ cancel, yielding the alternative form:

$$\boxed{D' = \frac{D_{\text{n}}D_{\text{p}}(n_0+p_0)}{D_{\text{n}}n_0+D_{\text{p}}p_0}}$$

The effective lifetime term τ' is given by:

$$\frac{1}{\tau'} = \frac{\mu_{\text{p}}p_0/\tau_{\text{n0}}+\mu_{\text{n}}n_0/\tau_{\text{p0}}}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}.$$

Under low injection, τ' reduces to the minority carrier lifetime, as shown below.

The final **ambipolar transport equation** is:

$$\boxed{\frac{\partial\delta\rho}{\partial t} = -\mu'\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x} + D'\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau'}}$$

Effective Ambipolar Parameters

For the case where both electrons and holes contribute significantly to conduction (i.e., in intrinsic or lightly doped material, or under high injection), the **ambipolar mobility** and **ambipolar diffusion coefficient** are given by:

$$\boxed{\mu' = \frac{(n_0 - p_0)\mu_\mathrm{n}\mu_\mathrm{p}}{n_0\mu_\mathrm{n} + p_0\mu_\mathrm{p}}}$$

$$\boxed{D' = \frac{D_\mathrm{n}D_\mathrm{p}(n_0 + p_0)}{D_\mathrm{n}n_0 + D_\mathrm{p}p_0}}$$

(Note: The sign in μ' indicates the drift direction of the ambipolar packet. For $n_0 > p_0$, $\mu' > 0$ and the packet drifts in the direction of the field, analogous to hole motion. The expression for D' is obtained by substituting the Einstein relation into the mobility-weighted form.)

The effective lifetime τ' depends on the recombination mechanism. Under low injection, it reduces to the minority carrier lifetime.

Important Simplifications for Practical Cases

Case 1: n-Type Material ($n_0 \gg p_0$, low injection $\delta\rho \ll n_0$)
For an n-type semiconductor, $n_0 \gg p_0$ and $\delta\rho \ll n_0$. Then:

$$\mu' \approx \frac{n_0\mu_\mathrm{n}\mu_\mathrm{p}}{n_0\mu_\mathrm{n}} = \mu_\mathrm{p}$$

$$D' \approx \frac{n_0 D_\mathrm{p}}{n_0} = D_\mathrm{p}$$

$$\frac{1}{\tau'} \approx \frac{\mu_\mathrm{n}n_0/\tau_{\mathrm{p}0}}{\mu_\mathrm{n}n_0} = \frac{1}{\tau_{\mathrm{p}0}}$$

Thus, the ambipolar equation reduces to the **hole diffusion equation**:

$$\boxed{\frac{\partial\delta\rho}{\partial t} = D_\mathrm{p}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{p}0}}}$$

Case 2: p-Type Material ($p_0 \gg n_0$, low injection $\delta\rho \ll p_0$)
Similarly, for p-type material:

$$\mu' \approx -\mu_\mathrm{n}, \qquad D' \approx D_\mathrm{n}, \qquad \tau' \approx \tau_{\mathrm{n}0}$$

The ambipolar equation reduces to the **electron diffusion equation**:

$$\boxed{\frac{\partial\delta\rho}{\partial t} = D_\mathrm{n}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{n}0}}}$$

Physical Interpretation of the Ambipolar Parameters

- **Ambipolar mobility** μ': In intrinsic material ($n_0 = p_0$), $\mu' = 0$ — there is no net drift of the excess packet in an applied field because electrons and holes drift in opposite directions, canceling each other's motion.
- **Ambipolar diffusion coefficient** D': This is a weighted average of D_n and D_p. The slower species (usually holes) limits the diffusion of the coupled pair.

- **Why $D' \approx D_{\mathrm{p}}$ in n-type**: The excess holes (minority carriers) control the diffusion rate because they are the limiting species. The abundant majority electrons quickly adjust to maintain neutrality.

Summary: The Ambipolar Approximation in a Nutshell

Key Takeaways

- The **internal electric field** $\mathcal{E}_{\mathrm{int}}$ couples electron and hole motion, forcing excess carriers to travel together
- This binding applies only to **excess carriers** (δn, δp), not total carriers
- The result is a single **ambipolar diffusion equation** for $\delta\rho = \delta n = \delta p$
- Under low injection in n-type material: $D' = D_{\mathrm{p}}$, $\mu' = \mu_{\mathrm{p}}$ (hole diffusion controls)
- Under low injection in p-type material: $D' = D_{\mathrm{n}}$, $\mu' = -\mu_{\mathrm{n}}$ (electron diffusion controls)
- In intrinsic material, the ambipolar mobility $\mu' = 0$ — the excess packet does not drift

Why Not Simply Set $\delta n = \delta p$ Exactly?

Recall our earlier inconsistency: setting $\delta n = \delta p$ exactly led to contradictory equations because $\mu_{\mathrm{n}} \neq \mu_{\mathrm{p}}$. The ambipolar approximation resolves this by:

1. Acknowledging that $\delta n \approx \delta p$ (quasi-neutrality), but not exactly equal
2. The small difference $\delta p - \delta n$ generates $\mathcal{E}_{\mathrm{int}}$ via Poisson's equation
3. This $\mathcal{E}_{\mathrm{int}}$ couples the equations and allows a single equation to emerge
4. The factor e/ϵ can amplify tiny charge imbalances into significant fields

Thus, the ambipolar approximation is **self-consistent**: the internal field adjusts to make the two species move together, and we describe their coupled motion with effective parameters.

Final Ambipolar Equation for Minority Carrier Diffusion

For most practical device analysis (p-n junctions, BJTs, solar cells), we work in the low-injection regime where the minority carrier diffusion equation applies:

$$\boxed{\frac{\partial \delta\rho}{\partial t} = D_{\mathrm{minority}} \frac{\partial^2 \delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{minority}}}}$$

where:

- For n-type: $D_{\mathrm{minority}} = D_{\mathrm{p}}$, $\tau_{\mathrm{minority}} = \tau_{\mathrm{p0}}$, $\delta\rho =$ excess holes
- For p-type: $D_{\mathrm{minority}} = D_{\mathrm{n}}$, $\tau_{\mathrm{minority}} = \tau_{\mathrm{n0}}$, $\delta\rho =$ excess electrons

This equation forms the foundation for analyzing all bipolar devices, as we will see in subsequent lectures.

Conceptual Clarification: How Can Electrons and Holes Travel Together Without Annihilating?

A common conceptual hurdle arises when students compare two seemingly contradictory statements:

1. In the ambipolar transport model, excess electrons and holes travel together as a coupled pair, maintaining approximate local charge neutrality ($\delta n \approx \delta p$) at every point in space and time.
2. In the recombination dynamics analysis, we derived that the excess carrier recombination rate is $r = \delta n/\tau_{n0}$ (or $r = \delta p/\tau_{p0}$), meaning that when an electron and a hole meet, they annihilate.

The natural question arises: **If electrons and holes are traveling together, why don't they instantly recombine and vanish?**

The Resolution: Two Vastly Different Time Scales The key to resolving this apparent paradox lies in recognizing that there are **two completely distinct time scales** at play in semiconductor physics:

Parameter	Symbol	Typical Magnitude	Physical Meaning
Momentum relaxation time	τ_n, τ_p	10^{-13} – 10^{-12} s	Time between scattering events (collisions with lattice, impurities)
Carrier lifetime	τ_{n0}, τ_{p0}	10^{-9} – 10^{-3} s	Time before an electron and hole recombine

The critical observation is that:

$$\boxed{\tau_{n0}, \tau_{p0} \gg \tau_n, \tau_p} \tag{48}$$

Typically, the carrier lifetime is **orders of magnitude larger** than the momentum relaxation time. For silicon at room temperature:

- $\tau_n \approx \tau_p \approx 10^{-13}$ s (picosecond scale)
- $\tau_{n0} \approx 10^{-6} - 10^{-3}$ s (microsecond to millisecond scale)

The ratio τ_{n0}/τ_n can be as large as 10^7 – 10^{10}!

What This Means Physically

Analogy: Two People Walking Together in a Crowded Room

Imagine two people (an electron and a hole) trying to find each other in a vast, crowded stadium filled with thousands of other people (lattice atoms, impurities, other carriers).

- **Momentum relaxation (scattering)**: Each person constantly bumps into other people in the crowd, changing direction every fraction of a second. This is like τ_n and τ_p — frequent, random collisions that randomize velocity.
- **Recombination (annihilation)**: The two specific people (electron and hole) are searching for each other to "annihilate" (recombine). Even if they are walking in the same direction, the stadium is so crowded and large that it takes a **very long time** (millions of collisions) before they actually meet.
- The **lifetime** τ_{n0} is the average time before this meeting happens.

Why the Internal Field Binds Without Causing Recombination The internal electric field $\mathcal{E}_{\text{int}}$ that couples the excess electron and hole pulses arises from **electrostatic attraction** — it is a long-range Coulomb force that acts continuously over large distances. This force ensures that the *centers* of the excess carrier pulses move together.

However, recombination requires a **close encounter** — the electron and hole must come within a very short distance (typically a few nanometers, comparable to the exciton Bohr radius) to recombine. The probability of such a close encounter at any given moment is extremely small.

Thus:

- The **internal field** ensures the pulses drift and diffuse together on average
- **Recombination** is a rare event that occurs when, by chance, a specific electron and hole come very close
- Because $\tau_{\text{n0}} \gg \tau_{\text{n}}$, the carriers undergo **billions of scattering events** before finally recombining

Mathematical Justification We can make this argument more quantitative. Consider an n-type semiconductor under low injection. The excess holes (minority carriers) have:

- Mean time between scattering events: $\tau_{\text{p}} \approx 10^{-13}$ s
- Mean time before recombination: $\tau_{\text{p0}} \approx 10^{-6}$ s

The number of scattering events a typical hole experiences before recombining is:

$$N_{\text{scat}} = \frac{\tau_{\text{p0}}}{\tau_{\text{p}}} \approx \frac{10^{-6}}{10^{-13}} = 10^{7}$$

The hole undergoes **ten million collisions** with the lattice and impurities before finally encountering an electron to recombine. During this time, it can travel a significant distance (the diffusion length $L_p = \sqrt{D_p \tau_{\text{p0}}}$, typically tens of micrometers).

The Role of Charge Neutrality Approximate local charge neutrality ($\delta n \approx \delta p$) does **not** mean that every electron is physically coincident with a hole. Rather, it means that in any small volume (large enough to contain many carriers but small compared to device dimensions), the *total number* of excess electrons approximately equals the *total number* of excess holes.

Within that volume, individual electrons and holes are moving randomly, scattering frequently. The probability that any specific electron-hole pair comes close enough to recombine during a given short time interval is tiny. Only after many scattering events (on the time scale of τ_{n0} or τ_{p0}) does recombination eventually occur.

Key Takeaways

- The **internal electric field** couples the *centers of mass* of the excess electron and hole distributions via long-range Coulomb forces
- **Recombination** requires a short-range encounter (within nanometers) and is therefore a much rarer event
- The vast separation of time scales ($\tau_{\text{n0}}, \tau_{\text{p0}} \gg \tau_{\text{n}}, \tau_{\text{p}}$) allows the ambipolar transport approximation to be valid: excess carriers can travel together for long distances and times before eventually recombining
- Approximate local charge neutrality ($\delta n \approx \delta p$) refers to *average* concentrations over volumes containing many carriers, not to individual electron-hole pairs

Connection to the Drift-Diffusion-Poisson Model In the DDP model, the continuity equations track *concentrations* (averages over many carriers), not individual particles. The recombination terms $-\delta n/\tau_{\mathrm{n0}}$ and $-\delta p/\tau_{\mathrm{p0}}$ represent the *average rate* at which carriers disappear from the ensemble. This average rate is small precisely because τ_{n0} and τ_{p0} are large compared to the scattering time.

Thus, there is no contradiction:

- At the **macroscopic (continuum)** level, we describe coupled drift-diffusion of excess electron and hole *populations*
- At the **microscopic (particle)** level, individual electrons and holes undergo frequent scattering but rarely recombine
- The internal field ensures the *average* motions are correlated, while recombination represents the slow, statistical annihilation of the populations

This separation of time scales is what makes semiconductor devices possible: carriers can travel significant distances (across a p-n junction, through a transistor base) before recombining, allowing for current flow and amplification.

12.5 Applications of the Ambipolar Transport Equation

We now solve the ambipolar transport equation for several important problems. These examples illustrate the behavior of excess carriers in semiconductor materials, and the results will be used later in the discussion of p-n junctions and other semiconductor devices.

The following examples use several common simplifications in the solution of the ambipolar transport equation. Table 1 summarizes these simplifications and their effects.

Specification	**Effect**
Steady state	$\frac{\partial(\delta n)}{\partial t} = 0, \quad \frac{\partial(\delta p)}{\partial t} = 0$
Uniform distribution of excess carriers (uniform generation rate)	$\frac{\partial^2(\delta n)}{\partial x^2} = 0, \quad \frac{\partial^2(\delta p)}{\partial x^2} = 0$
Zero electric field	$\mathcal{E} = 0, \quad \mathcal{E}\frac{\partial(\delta n)}{\partial x} = 0, \quad \mathcal{E}\frac{\partial(\delta p)}{\partial x} = 0$
No excess carrier generation	$g = 0$
No excess carrier recombination (infinite lifetime)	$\frac{\delta n}{\tau_{n0}} = 0, \quad \frac{\delta p}{\tau_{p0}} = 0$

Example 6.2: Time Decay of Excess Carriers (Recombination)

Objective: Determine the time behavior of excess carriers as a semiconductor returns to thermal equilibrium.

Consider an infinitely large, homogeneous n-type semiconductor with zero applied electric field. Assume that at time $t = 0$, a uniform concentration of excess carriers exists in the crystal, but assume that $g = 0$ for

$t > 0$. If we assume that the concentration of excess carriers is much smaller than the thermal-equilibrium electron concentration, then the low-injection condition applies. Calculate the excess carrier concentration as a function of time for $t \geq 0$.

Solution:

For the n-type semiconductor, we need to consider the ambipolar transport equation for the minority carrier holes, which is:

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p \mathcal{E} \frac{\partial(\delta p)}{\partial x} + g - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

We are assuming a uniform concentration of excess holes so that:

$$\frac{\partial^2(\delta p)}{\partial x^2} = \frac{\partial(\delta p)}{\partial x} = 0$$

For $t > 0$ we are also assuming that $g = 0$. The equation reduces to:

$$\frac{d(\delta p)}{dt} = -\frac{\delta p}{\tau_{p0}}$$

Since there is no spatial variation, the total time derivative may be used. At low injection, the minority carrier hole lifetime τ_{p0} is a constant. The left-hand side is the time rate of change of δp and the right-hand side is the recombination rate. The solution is:

$$\delta p(t) = \delta p(0)\, e^{-t/\tau_{p0}}$$

where $\delta p(0)$ is the uniform concentration of excess carriers that exists at time $t = 0$. The concentration of excess holes decays exponentially with time, with a time constant equal to the minority carrier hole lifetime.

From the charge neutrality condition, we have $\delta n = \delta p$, so the excess electron concentration is given by:

$$\delta n(t) = \delta p(t) = \delta p(0)\, e^{-t/\tau_{p0}}$$

Comment: The excess electrons and holes recombine at the rate determined by the excess minority carrier hole lifetime in the n-type semiconductor.

Example 6.3: Time-Dependent Buildup to Steady State

Objective: Determine the time dependence of excess carriers in reaching a steady-state condition.

Again consider an infinitely large, homogeneous n-type semiconductor with zero applied electric field. Assume that for $t < 0$, the semiconductor is in thermal equilibrium and that for $t \geq 0$, a uniform generation rate exists in the crystal. Calculate the excess carrier concentration as a function of time assuming the condition of low injection.

Solution:

The condition of a uniform generation rate and a homogeneous semiconductor again implies that:

$$\frac{\partial^2(\delta p)}{\partial x^2} = \frac{\partial(\delta p)}{\partial x} = 0$$

The ambipolar transport equation for minority carrier holes reduces to:

$$g - \frac{\delta p}{\tau_{p0}} = \frac{d(\delta p)}{dt}$$

This is a first-order differential equation. To solve it, rearrange:

$$\frac{d(\delta p)}{dt} + \frac{\delta p}{\tau_{p0}} = g$$

The solution consists of a homogeneous part plus a particular solution. The homogeneous solution is $\delta p_h = Ae^{-t/\tau_{p0}}$. The particular solution (steady state) is found by setting $d(\delta p)/dt = 0$, giving $\delta p_{ss} = g\tau_{p0}$. The complete solution is:

$$\delta p(t) = g\tau_{p0} + Ae^{-t/\tau_{p0}}$$

The constant A is determined from the initial condition. At $t = 0$, $\delta p(0) = 0$ (thermal equilibrium). Therefore:

$$0 = g\tau_{p0} + A \quad \Rightarrow \quad A = -g\tau_{p0}$$

Thus:

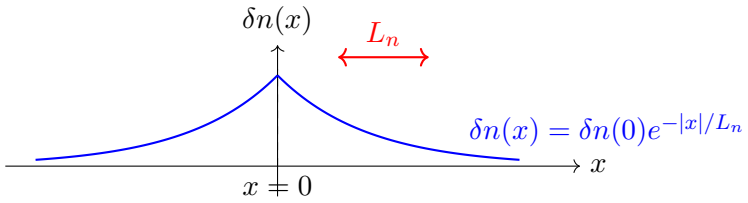
$$\boxed{\delta p(t) = g\tau_{p0}\left(1 - e^{-t/\tau_{p0}}\right)}$$

Comment: We may note that as $t \to \infty$, a steady-state excess hole and electron concentration of $g\tau_{p0}$ is reached. The excess minority carrier hole concentration increases with time with the same time constant τ_{p0}, which is the excess minority carrier lifetime. The excess carrier concentration reaches a steady-state value as time goes to infinity, even though a steady-state generation of excess electrons and holes exists. This steady-state effect can be seen by setting $d(\delta p)/dt = 0$ in the original equation, which yields $g = \delta p/\tau_{p0}$, i.e., generation rate equals recombination rate.

Example 6.4: Steady-State Spatial Decay (Diffusion Length)

Objective: Determine the steady-state spatial dependence of the excess carrier concentration.

Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will begin diffusing in both the $+x$ and $-x$ directions. Calculate the steady-state excess carrier concentration as a function of x.



Solution:

The ambipolar transport equation for excess minority carrier electrons in p-type material is:

$$D_n\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n\mathcal{E}\frac{\partial(\delta n)}{\partial x} + g - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

From our assumptions:

- $\mathcal{E} = 0$ (zero applied field)
- $g = 0$ for $x \neq 0$ (generation only at $x = 0$)
- $\partial(\delta n)/\partial t = 0$ (steady state)

Assuming a one-dimensional crystal, the equation reduces to:

$$D_n\frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{\tau_{n0}} = 0$$

Dividing by the diffusion coefficient:

$$\frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{D_n \tau_{n0}} = \frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{L_n^2} = 0$$

where we have defined $L_n^2 = D_n \tau_{n0}$. The parameter L_n has units of length and is called the **minority carrier electron diffusion length**.

The general solution to this differential equation is:

$$\delta n(x) = Ae^{-x/L_n} + Be^{x/L_n}$$

As the minority carrier electrons diffuse away from $x = 0$, they recombine with the majority carrier holes. The minority carrier electron concentration will then decay toward zero at both $x = +\infty$ and $x = -\infty$. These boundary conditions mean that $B = 0$ for $x > 0$ and $A = 0$ for $x < 0$.

The solution may then be written as:

$$\delta n(x) = \delta n(0)\, e^{-x/L_n} \quad \text{for } x \geq 0$$
$$\delta n(x) = \delta n(0)\, e^{+x/L_n} \quad \text{for } x \leq 0$$

or more compactly:

$$\boxed{\delta n(x) = \delta n(0)\, e^{-|x|/L_n}}$$

where $\delta n(0)$ is the value of the excess electron concentration at $x = 0$.

Comment: The steady-state excess concentration decays to $1/e$ of its value at $x = \pm L_n$. The diffusion length $L_n = \sqrt{D_n \tau_{n0}}$ represents the average distance a minority carrier diffuses before recombining.

From charge neutrality, $\delta p(x) = \delta n(x)$, so the steady-state excess majority carrier hole concentration also decays exponentially with distance with the same characteristic minority carrier electron diffusion length L_n.

Under low injection ($\delta n(0) \ll p_0$), the total concentration of majority carrier holes barely changes. However, the minority carrier concentration may change by many orders of magnitude.

Typical values for silicon at 300 K:

- For p-type Si with $N_a = 10^{16}$ cm^{-3}: $D_n \approx 25$ cm^2/s, $\tau_{n0} \approx 10^{-6}$ s $\Rightarrow L_n \approx 50$ μm
- For n-type Si with $N_d = 10^{16}$ cm^{-3}: $D_p \approx 10$ cm^2/s, $\tau_{p0} \approx 10^{-6}$ s $\Rightarrow L_p \approx 32$ μm